

Probabilistic Assessment and Forecasting of Landslide Occurrences using Machine Learning

Objective of my component – My objective is creating a ML driven system to early identification of landslide before it strike harder to the country

Multi-Factor Probability Calculation: Establishing a highly accurate, real-time probability engine that calculates landslide risks by mathematically correlating dynamic rainfall volumes with static slope geometry and soil saturation variables simultaneously.

https://www.researchgate.net/publication/366114343_Geoinformatics_approach_to_classifying_landslide_hazard_in_The_Hantana_Mountain_and_its_vicinity_in_Sri_Lanka

Inclusive and Localized Live Mapping: Designing a Web-GIS live map featuring high-contrast accessible user interface themes for elderly citizens, alongside dual-language localization in native languages for village-level alerts.

[Accessibility essentials for GIS and mapping](#)

<https://lrip.nbro.gov.lk/portal/apps/webappviewer/index.html?id=7524ad70c98e4124a6f4dfed6ea615fe>

Explainable Micro-Level Terrain Failures: Transitioning from broad regional warnings to site-specific evaluations by integrating an Explainable AI layer to show the exact percentage breakdown of the environmental and human causes behind a predicted slope failure.

https://www.dmc.gov.lk/index.php?option=com_dmcreports&view=reports&Itemid=276&report_type_id=5&lang=en

https://www.dmc.gov.lk/images/dmcreports/Landslide_Early_warning_report_at_1600hrs_on_13_1781348104.pdf

For the static human factors (road cuts and building loads), I am digitizing a specific high-risk pilot area (e.g., a section of the Badulla or Kegalle district) using Department of Census and Statistics shapefiles and OpenStreetMap data. For dynamic factors, I am using interpolation algorithms on nearby NBRO rain gauge data.

Micro-Level Terrain Failure & Explainable AI (XAI)

To move away from regional alerts to micro-level site-specific evaluation, you must couple machine learning with feature attribution.

- Machine Learning Framework:
 - XGBoost / LightGBM: Highly accurate for tabular spatial data (combining static road cuts, building loads, soil type, and dynamic rainfall). They handle missing data well and natively support spatial features.
- Explainable AI Layer:
 - SHAP (SHapley Additive exPlanations): The gold standard for your exact requirement. It calculates the exact percentage breakdown or impact vector of environmental (rainfall, slope) and human causes (road cuts, building loads) behind each specific prediction.
 - LIME (Local Interpretable Model-agnostic Explanations): Useful as a faster, localized alternative to explain individual grid-cell slope failures.

Data Ingestion & Spatial Digitization (GIS Pipeline)

You need tools to clean, store, and manipulate the [Department of Census and Statistics](#) shapefiles, OpenStreetMap (OSM) human features, and [Hantana/Kegalle/Badulla](#) terrain data.

- GIS Desktop (Digitization & Processing):
 - QGIS: Free, open-source, and perfect for digitizing building loads, extracting road cuts from OSM data, and clipping shapefiles to your Badulla or Kegalle pilot zones.
- Spatial Database:
 - PostgreSQL + PostGIS extension: Crucial for real-time engines. It allows you to run fast spatial queries (e.g., finding the distance of a slope to the nearest road cut) using SQL.
- OSM Data Extraction:
 - OSMnx: A Python library to directly download, model, and analyze road networks and building footprints from OpenStreetMap for your pilot area.

Inclusive & Localized Live Web-GIS Map (Frontend)

Your web interface needs to handle live geospatial vector data, support accessibility (WCAG), and offer dual-language localization (Sinhala/Tamil) for village-level alerts.

- Mapping Library:

- MapLibre GL JS / Mapbox GL JS: Uses WebGL for smooth, hardware-accelerated rendering of real-time hazard layers down to the micro-level grid.
 - Leaflet: A lightweight alternative if your map elements are simple vectors, highly compatible with older mobile devices used in villages.
- Frontend Framework:
 - React.js or Vue.js: Excellent for managing complex application states like toggling high-contrast themes and switching languages instantly.
- Accessibility & UI Themes (Elderly Care):
 - Tailwind CSS: Allows you to easily implement high-contrast modes, scalable font sizes (rem units), and accessible focus states.
 - Radix UI / Headless UI: Provides fully accessible (WCAG 2.1 compliant) unstyled components (modals, dropdowns) that work perfectly with screen readers.
- Localization (Dual-Language Alerts):
 - i18next / react-i18next: The industry standard for handling multi-language translation keys (English, Sinhala, Tamil). It lets you dynamically translate village-level hazard warnings based on user preference.